



ECEN 314 – Signals and Systems

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Module 5 - Convolution

Learning Objectives



- Be able to form the convolution of two signals.

Convolution



Convolution is an operation involving two signals. For two waveforms, $x(t)$ and $y(t)$, their convolution is given by

$$z(t) = x(t) * y(t) = \int_{-\infty}^{\infty} x(v)y(t-v)dv.$$

Note that using a change of variables, $w = t - v$ this can be rewritten as

$$\begin{aligned} z(t) &= - \int_{\infty}^{-\infty} x(t-w)y(w)dw \\ &= \int_{-\infty}^{\infty} x(t-w)y(w)dw. \end{aligned}$$

Therefore $x(t) * y(t) = y(t) * x(t)$.

In general, convolution is a rather nasty operation and we often try to find various tricks to avoid performing convolution, but sometimes we just have no choice, so the next several slides will go through numerous examples of computing convolutions.

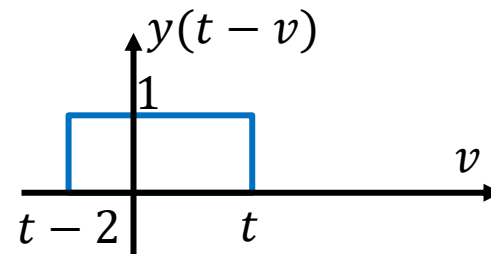
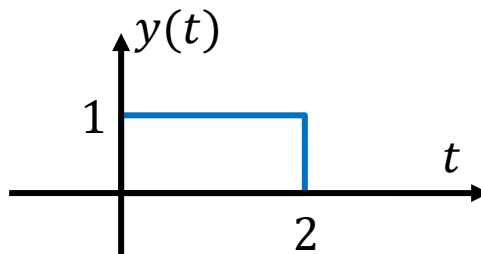
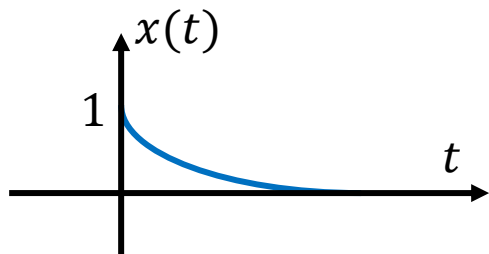


Convolution



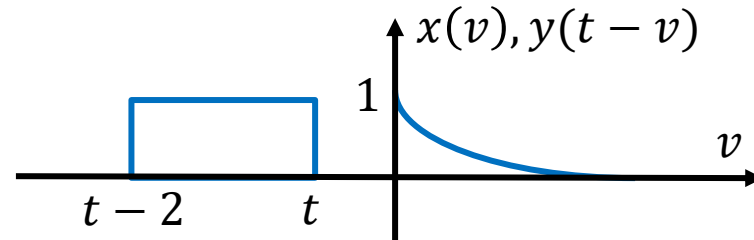
Example 5.1: Suppose $x(t) = e^{-t}u(t)$ and $y(t) = \text{rect}\left(\frac{t-1}{2}\right)$.

Find $z(t) = x(t) * y(t) = \int_{-\infty}^{\infty} x(v)y(t-v)dv$.



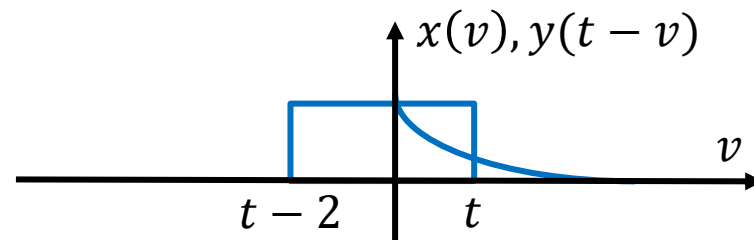
Case I: $t < 0$,

Since the two waveforms do not overlap, the product $x(v)y(t-v) = 0$ for all u and therefore $z(t) = 0$.



Case II: $0 < t < 2$,

$$z(t) = \int_0^t e^{-v} dv = -e^{-v} \Big|_0^t = 1 - e^{-t}$$

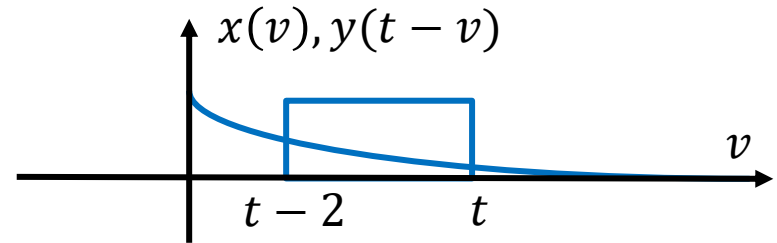


Convolution



Case III: $t > 2$,

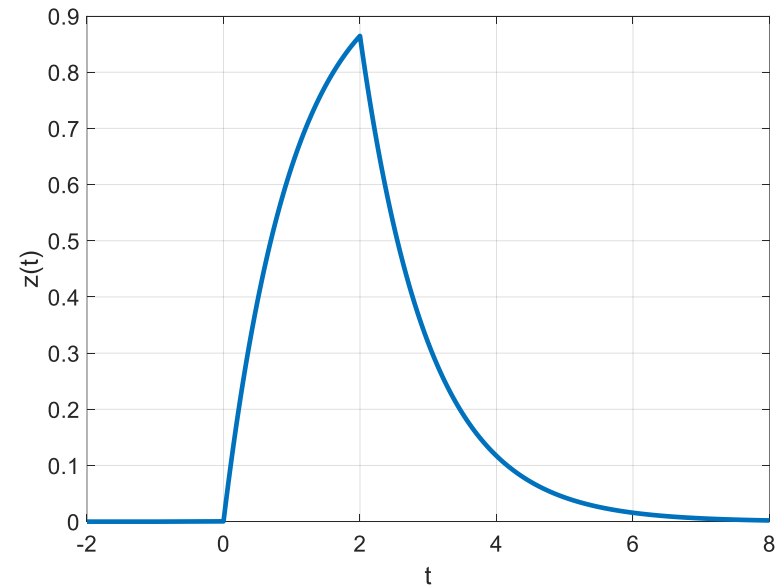
$$\begin{aligned} z(t) &= \int_{t-2}^t e^{-v} dv = -e^{-v} \Big|_{t-2}^t \\ &= e^{-(t-2)} - e^{-t} = (e^2 - 1)e^{-t}, \\ &\quad \text{or equivalently,} \\ z(t) &= (1 - e^{-2})e^{-(t-2)}. \end{aligned}$$



Putting these three results together, we get:

$$z(t) = x(t) * y(t) = \begin{cases} 0, & t < 0, \\ 1 - e^{-t}, & 0 < t < 2, \\ (e^2 - 1)e^{-t}, & t > 2. \end{cases}$$

This procedure of breaking the convolution integral into different cases depending on the values of t is pretty common.

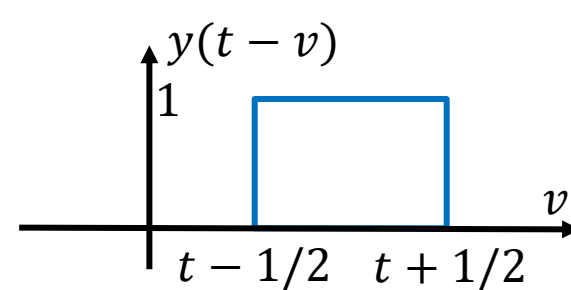
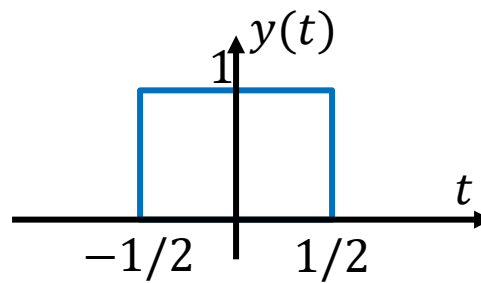
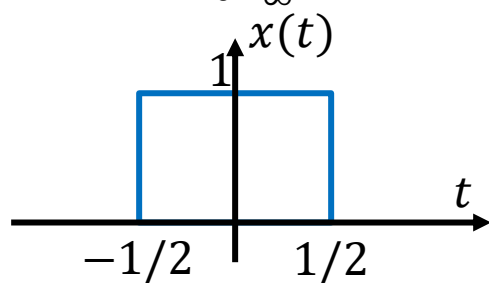


Convolution

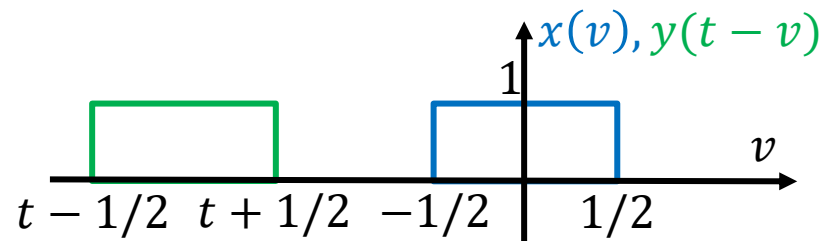


Example 5.2: Sometimes it is possible to evaluate a convolution geometrically and thereby avoid using calculus. Suppose $x(t) = y(t) = \text{rect}(t)$.

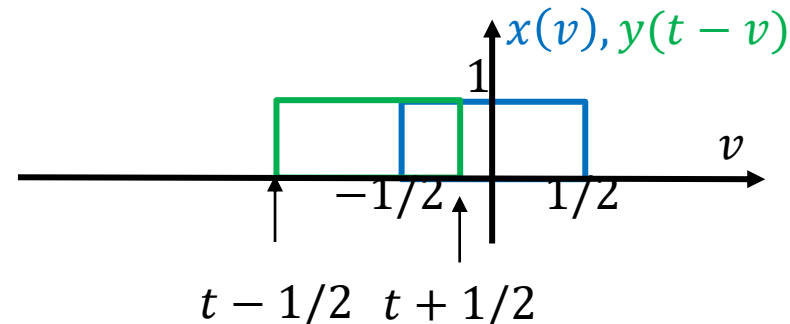
Find $z(t) = \int_{-\infty}^{\infty} x(v)y(t-v)dv$



Case I: $t < -1$,
 $z(t) = 0$.



Case II: $-1 < t < 0$,
 $z(t) = \int_{-1/2}^{t+1/2} du = t + 1$.

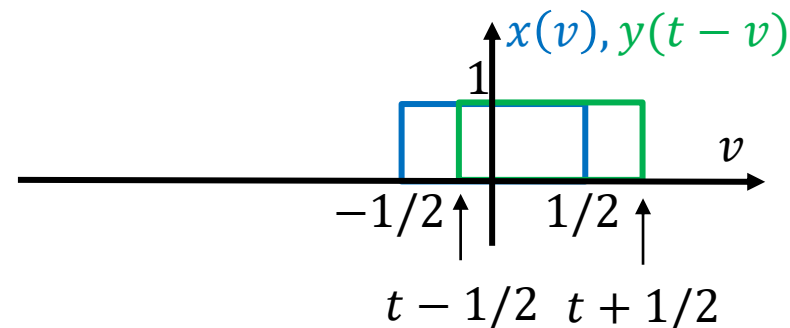


Convolution



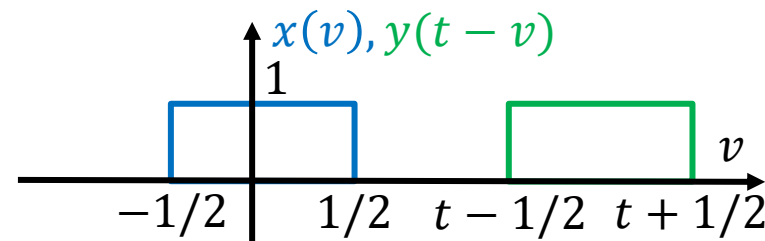
Case III: $0 < t < 1$,

$$z(t) = \int_{t-1/2}^{1/2} du = -t + 1.$$



Case IV: $t > 1$,

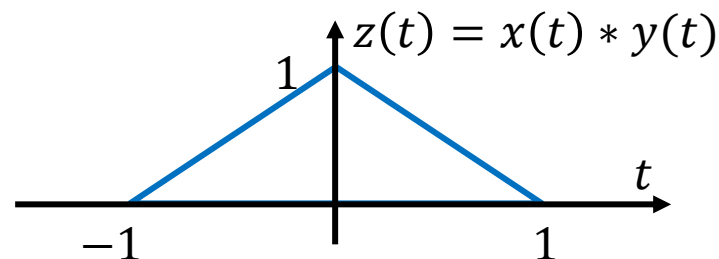
$$z(t) = 0.$$



Putting these four results together, we get:

$$z(t) = x(t) * y(t) = \begin{cases} 0, & |t| > 1, \\ 1 - |t|, & |t| < 1. \end{cases}$$

Alternatively, $z(t) = \text{tri}(t)$.



Convolution



Example 5.3: Suppose $x(t) = y(t) = e^{-t}u(t)$.
Find the convolution $z(t) = x(t) * y(t)$.

Convolution



Example 5.4: Suppose $x(t) = e^{-t}u(t)$ and $y(t) = e^{-2t}u(t)$.
Find the convolution $z(t) = x(t) * y(t)$.

Convolution



Example 5.5: Suppose $x(t) = \sin(\omega t)$ and $y(t) = e^{-2t}u(t)$.
Find the convolution $z(t) = x(t) * y(t)$.

Properties of Convolution

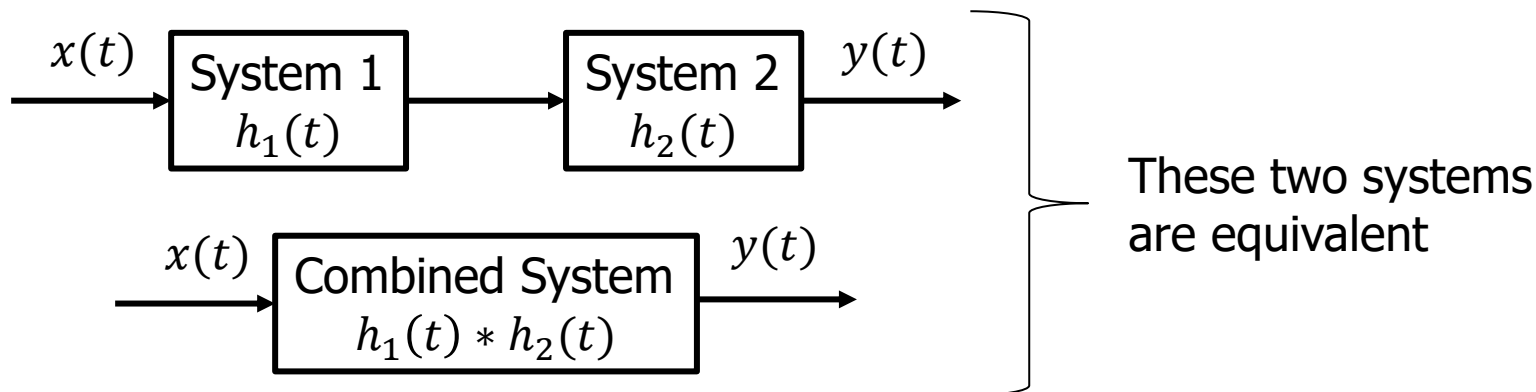


Now that we have a handle on how to perform convolutions, let's look at some general properties of convolution and their implication on LTI systems.

1. Commutative Property - $x(t) * y(t) = y(t) * x(t)$.

→ When performing convolution, we can “flip + shift” either signal.

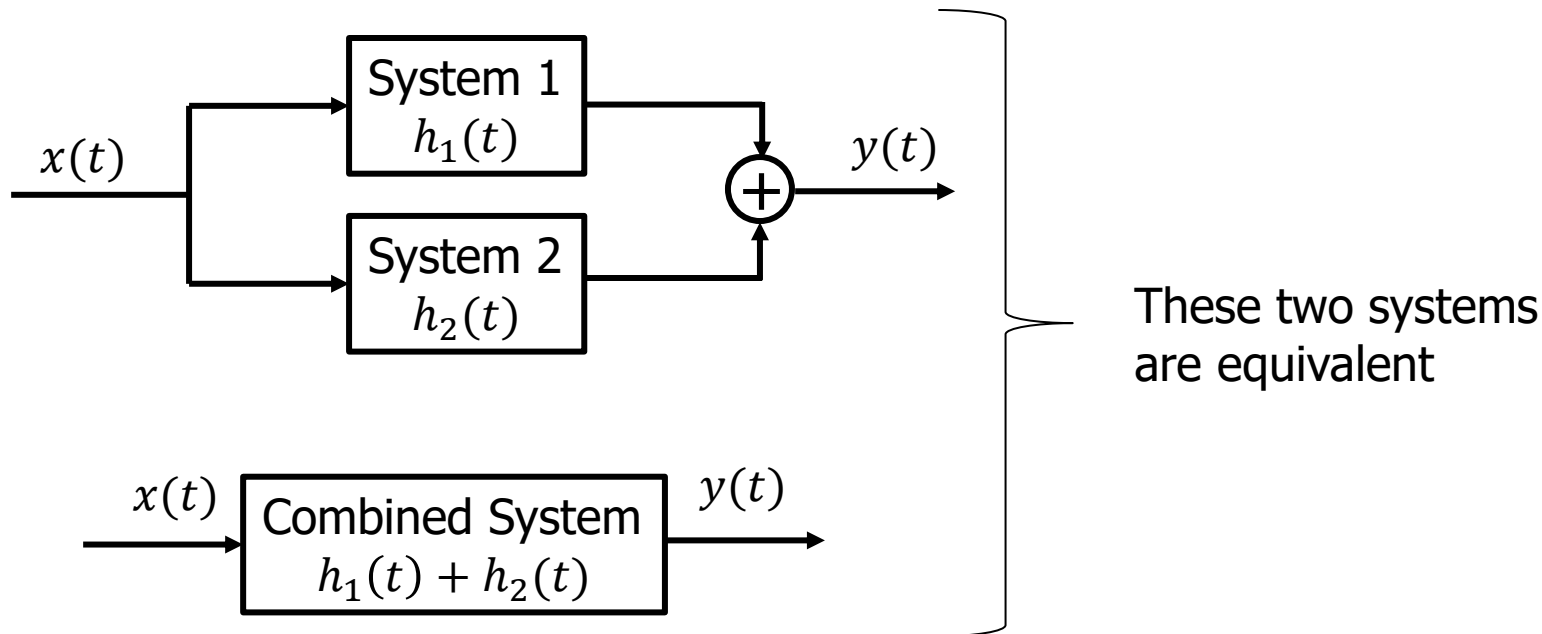
2. Associative Property – $[x(t) * y(t)] * z(t) = x(t) * [y(t) * z(t)]$.



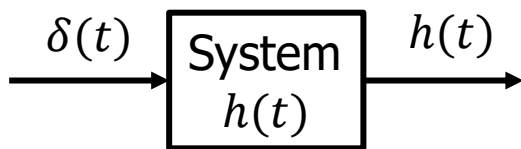
Properties of Convolution



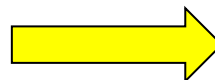
3. Distributive Property - $x(t) * [y(t) + z(t)] = x(t) * y(t) + x(t) * z(t)$.



4. Identity - $x(t) * \delta(t) = \delta(t) * x(t) = x(t)$



$$h(t) = \delta(t) * h(t)$$



When it comes time to compute convolutions, the delta function is your new best friend.

Properties of Convolution



Example 5.6: Suppose a certain LTI system has an impulse response of $h(t) = \text{rect}\left(\frac{t-1}{2}\right)$. Find the output of that system due to each of the following inputs:

(a) $x_1(t) = \text{rect}\left(t - \frac{1}{2}\right)$,

(b) $x_2(t) = \delta(t)$,

(c) $x_3(t) = \delta(t) + \text{rect}\left(t - \frac{1}{2}\right)$.

Convolution in Discrete Time



Example 5.7: Suppose a certain discrete-time LTI system has an impulse response of $h[n] = u[n]$. Find the output of that system if the input is $x[n] = \alpha^n u[n]$ (assuming $|\alpha| < 1$).